

Exercise for Sect. 2.8

Exercise

2.8.1

Is the hypothesis that the sequence (a_n) be bounded necessary in Lemma 2.41.

Proof:

(2) $\Rightarrow$ (1) **fails** Let $J = \{k^2 : k \in \mathbb{N}\}$ is density zero, and $a_n = n$ for $n \in J$ otherwise 0.

The sequence converges to 0 for $n \notin J$ but the subsequence

$$A_{n^2+1} = \frac{1}{n^2+1} \sum_{j=0}^{n^2} a_j = \frac{1}{n^2+1} \frac{n(n+1)(2n+1)}{6} \rightarrow \infty.$$

Therefore, $\lim_{n \rightarrow \infty} \frac{1}{n} \sum a_j \neq 0$.

(2) $\Rightarrow$ (3) **fails** Same as above.

(1) $\Rightarrow$ (3) **fails** J is same as above but $a_n = n^{\frac{1}{4}}$ for $n \in J$. Then the first $m^2 + 1$

summation is $A_{m^2+1} = \frac{1}{m^2} \sum_{j=0}^{m^2+1} a_j = \frac{1}{m^2} \sum_{k=1}^m k^{\frac{1}{2}}$. The subsequence converges by integral test and it should converges to zero. Since the whole sequence is decreasing except $n \in J$, the sequence converges to zero.

But for squared sequence, $A_{m^2+1} = \frac{1}{m^2} \sum_{k=1}^m k = \frac{1}{m^2} \cdot \frac{m(m+1)}{2} \rightarrow \frac{1}{2}$. Therefore the whole sequence may not converge to zero.

(3) $\Rightarrow$ (1) **still true** By Cauchy-Schwarz inequality $\left(\frac{1}{n} \sum_{j=0}^{n-1} a_j\right)^2 \leq \frac{1}{n} \sum_{j=0}^{n-1} a_j^2$.

(1) $\Rightarrow$ (2) **still true** By the proof in textbook, it doesn't use the bounded condition.

(3) $\Rightarrow$ (1) **still true** Combine the two above.

□

Exercise

2.8.2

Give an alternative proof of $(1) \implies (5)$ by proving the following statements:

1. Any factor of a weak-mixing transformation is weak-mixing.
2. A complex-valued eigenfunction f of U_T has constant modulus.
3. If f is an eigenfunction of U_T , then $x \mapsto \arg\left(\frac{f(x)}{|f(x)|}\right)$ is a factor map from $(X, \mathcal{B}, \mu, T)$ to $(\mathbb{T}, \mathcal{B}_{\mathbb{T}}, m_{\mathbb{T}}, R_{\alpha})$ for some α .

Proof:

1. Suppose $(Y, \mathcal{B}_Y, \nu, S)$ is a factor of $(X, \mathcal{B}_X, \mu, T)$ with the factor map φ . Suppose $A, B \in \mathcal{B}_Y$, $A' = \varphi^{-1}(A)$, $B' = \varphi^{-1}(B)$ are in $\mathcal{B}_X$. By the assumption, $\frac{1}{N} \sum_{n=0}^{N-1} |\mu(A' \cap T^{-n}B') - \mu(A')\mu(B')| \rightarrow 0$. Then,

$$\frac{1}{N} \sum_{n=0}^{N-1} |\nu(A \cap S^{-n}B) - \nu(A)\nu(B)| = \frac{1}{N} \sum_{n=0}^{N-1} |\mu(\varphi^{-1}(A \cap S^{-n}B)) - \mu(\varphi^{-1}(A))\mu(\varphi^{-1}(B))| \rightarrow 0$$

by the measure-preserving property of φ .

2. Suppose $U_T f = \lambda f$ almost everywhere. Take norm on both sides and U_T is isometry, $\|f\|_2 = |\lambda| \|f\|_2$ implies $|\lambda| = 1$. Take absolute value pointwise on both sides yields $U_T |f(x)| = |U_T f(x)| = |\lambda| |f(x)| = |f(x)|$ almost everywhere. Thus, $|f|$ is T -invariant and it is constant by the ergodicity of T (from weak-mixing.)
- 3.

□

Exercise**2.8.3**

If a measure-preserving system $(Y, \mathcal{B}_Y, \nu, S)$ is not totally ergodic then there exists a measure-preserving system $(X, \mathcal{B}, \mu, T)$ and $K > 1$ with the property that $(Y, \mathcal{B}_Y, \nu, S)$ is measurably isomorphic to the system

$$(X^{(K)}, \mathcal{B}^{(K)}, \mu^{(K)}, T^{(K)}).$$

Proof:

□

Exercise**2.8.4**

For a measure-preserving system $(X, \mathcal{B}, \mu, T)$ and function $f \in L^2_\mu$, show that the function $n \mapsto \langle U_T^n f, f \rangle$ is positive-definite. Apply the Herglotz-Bochner theorem to translate the problem into one concerning a function on $\mathbb{S}^1$, and then use the fact that $\frac{1}{N} \sum_{n=1}^N \rho^n$ converges for $\rho \in \mathbb{S}^1$.

Proof:

□